

Qualification Pack



Pipe Fitter (Power Plant)

QP Code: PSS/Q0201

Version: 3.0

NSQF Level: 3

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi-110021 || email: bishwajeet.singh@psscindia.org



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PSS/Q0201: Pipe Fitter (Power Plant)

Brief Job Description

It involves marking out the pipes as per specifications, and then use hand tools, portable power tools, manually operated machine tools and shaping, fabricating, fitting and assembly techniques appropriate to the operations being performed.

Personal Attributes

The job requires the individual to have good physical strength, strong hands, ability to work for long working hours/nights, good eye visibility and ability to communicate to customer and resolve their problems. The individual should be ethical and well behaved.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N0201: Perform fitting and assembly operations on pipes to produce pipework systems](#)
2. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
3. [SGJ/N1702: Optimize resource utilization at workplace](#)
4. [PSS/N1331: Apply basic health and safety practices for power related work](#)
5. [PSS/N1336: Working effectively with others](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Generation
Occupation	Fitting and Assembly
Country	India
NSQF Level	3
Credits	13
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7136.30



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Minimum Educational Qualification & Experience	8th grade pass plus 2-year NTC plus 1 Year NAC (Relevant Field) OR 8th grade pass with 1 Year of experience Relevant Experience OR 5th grade pass with 4 Years of experience Relevant Experience
Minimum Level of Education for Training in School	Same as Min.Ed for Job
Pre-Requisite License or Training	ITI/other Training on pipe fabrication/ installation / commissioning. Pipe brazing/soldering/Welding and Oxy-Fuel Gas Cutting
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/03/2025
NSQC Approval Date	31/03/2022
Version	3.0
Reference code on NQR	2022/POW/PSSC/06921
NQR Version	3.0



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PSS/N0201: Perform fitting and assembly operations on pipes to produce pipework systems

Description

This unit covers the pipe fabrication, fitting and assembly operations on various types of pipes to produce pipework systems as per given specifications. The candidate will be expected to carry out the pipe fabrication, fitting and assembly activities with understanding of the types of equipment used, the manufacturing techniques, and the operating and safety procedures that are required.

Scope

The scope covers the following :

- Working safely
- Preparing for pipe fabrication, fitting or assembling operations
- Marking out the components
- Performing general pipe fitting operations
- Performing pipework systems assembling operations
- Measuring and checking pipework

Elements and Performance Criteria

Working Safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (ppe) and other relevant safety regulations while performing fabrication and fitting operations
- PC3.** ensure work area is clean and safe from hazards
Hazards: use of power tools, trailing leads or hoses, damaged or badly maintained tools and equipment; using files with damaged or poor fitting handles; using machine tools; misuses of tools; not following laid-down maintenance procedures; hazardous working condition e.g. pressurized line, high temperature area, inflammable fluid or gas lines, working at confined space or at height etc.
- PC4.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC5.** ensure that all machines and machine tools are secured at all times
- PC6.** work safely in trenches, confined spaces and excavated areas
- PC7.** observe safety measures while working on high pressure line/system (steam, compressed air, hydraulic etc.
- PC8.** follow warning and safety signs (danger, out of service, etc.) while working with energized system (electrical systems, steam & compressed air system etc.) including road safety signs

Preparing for pipe fabrication, fitting or assembling operations

To be competent, the user/individual on the job must be able to:

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- PC9.** determine job requirement from job specification documents obtained from valid sources
Job requirements: raw materials or components required (type, quality, quantity); dimensions; limits and tolerances; operations required (list, sequence and procedures where applicable); shape or profiles to be fabricated; cutting, bending for fabricated forms; instruments and tools to be used; interdependencies; timelines
Job specification documents: detailed component drawings; approved sketches/illustrations; national, international and organisational standards; reference tables and charts; fabrication/casting drawings
Valid source: job instruction sheet/job card; work drawings and instructions; planning documentation; quality control documents; operation sheets; process specifications; instructions from supervisor
- PC10.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC11.** obtain the appropriate equipment, parts and accessories for the pipe fitting or assembling operation
- PC12.** check that all measuring equipment is within calibration date
Measuring equipment: external/internal micrometers, vernier caliper, tri-square, combination squares, rules, squares, protractors, depth micrometers, depth verniers, feeler gauges, bore/hole gauges, radius/profile gauges, thread gauges, hardness tester

Marking out the components

To be competent, the user/individual on the job must be able to:

- PC13.** prepare/determine suitable datum from which to mark out
- PC14.** apply a marking medium to enhance clarity of the marking out
- PC15.** use an appropriate method of marking out
Marking out methods: e.g. direct marking using tapes and markers, set-outs of pipework using templates, producing set wires, set-outs of pipework onto floor
- PC16.** use a range of marking out equipment (e.g. rules, squares, scribes, vernier instruments)
Marking tools: rules/tapes, dividers/trammels, scribes, punches, scribing blocks, squares, protractor, permanent markers
- PC17.** mark out a range of features
Features: datum lines; cutting guidelines; square and rectangular profiles; circular and radial profiles; angles; holes linearly positioned, boxed and on pitch circles
- PC18.** plan the pipe fitting activities before starting

Performing pipe fitting operations

To be competent, the user/individual on the job must be able to:

- PC19.** cut the pipes to the appropriate lengths making allowances for bending using appropriate cutting operations and techniques
Pipe cutting operations: cutting pipes to length with appropriate allowance for fittings, removing all external and internal burrs, cleaning pipe ends for soldering or cementing (where appropriate) or prior to aligning pipe for welding, the ends of the pipe shall be bevelled, cutting threads on pipe ends to the appropriate length (where appropriate), checking that prepared pipes are the correct length
Pipe cutting techniques: cutting out the rough profile using saws (e.g. hacksaw, chop saw, power saw), or oxy-fuel gas cutting torch or using pipe/tube cutter cutting a pipe thread (e.g. tapping or dieing), de-burring reamers, rubbing with abrasive cloth, wire pipe cleaning, filing (flat, square, curved), drilling holes



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- PC20.** produce pipework bends using the appropriate tools and equipment for the types and sizes of pipe
Pipe bending tools and equipment: hand operated pipe bender, bending springs, pipe expander, swaging kit, hydraulic pipe bending equipment, heating methods, fillers
Pipework bends and forms: angular bends, offsets, bridge sets, radii, internal, swaged ends, expansion loops, external swaged ends

Performing assembling operations

To be competent, the user/individual on the job must be able to:

- PC21.** assemble and secure the pipework as per job specifications using appropriate pipe assembly methods and techniques
pipe assembly methods: securing Pipework supports to structures, connecting pipe to equipment, fitting pipework supports, using gaskets, seals/sealing tapes or jointing compounds, connecting pipe-to-pipe, alignment/levelling equipment
Techniques: compression fittings, soldered fittings, cemented fittings, snap-on/push fittings, brazed fittings, welded joints, screwed connections
- PC22.** produce pipework assemblies which combine a range of different fittings
Pipe fittings: straight couplings, elbows, tee pieces, flanges, reduction pieces, drain/bleeding devices, unions
- PC23.** dismantle pipework assemblies without damage to components and/or subassemblies
Methods to dismantle: procedure for isolation and locking off a device/system; sequence of operations used to dismantle a device/system; proof marking, correct storage procedures for removed parts; release of pressure/force; extraction
- PC24.** deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve
- PC25.** keep the work area in a safe and tidy condition during and on completion of the manufacturing activities
- PC26.** return all tools and equipment to the correct location on completion of the fitting activities

Measuring and checking component

To be competent, the user/individual on the job must be able to:

- PC27.** perform the necessary checks for correct pipework assembly and dimensional accuracy
Checks: hydraulic pressure testing, gas/air leakage test, water leakage testing
Dimensions: linear dimensions (e.g. lengths, depths), diameters (e.g. external, internal), flatness, squareness, angles, profiles, hole size and position, thread size and fit
- PC28.** use the appropriate measuring equipment for checking activities
- PC29.** produce components within all of the applying standards
Components quality standards: pipes are bent to the appropriate shape/form and position, all pipe bends are free from buckling or deformation, appropriate fittings are used, and are secure and leak free, soldered and cemented fittings are free from excessive residues, the completed assembly meets the specific system requirements
- PC30.** generate stage inspection reports.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place



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- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** the Hazards associated with the pipe fitting activities hazards: e.g. handling long pipe lengths, using damaged or badly maintained tools and equipment, using pipe bending equipment, using heating and soldering equipment, using adhesives), and how they can be minimized, etc.
- KU11.** how to extract and use information from engineering drawings and related specifications in relation to work undertaken with the help of supervisor/ engineer when needed
- KU12.** how to interpret isometric drawings, imperial and metric systems of measurement, workpiece reference points and system of tolerancing (geometric dimensioning and tolerancing -- GD&T)
- KU13.** preparation of materials in readiness for the marking out activities, in order to enhance clarity, accuracy and safety
- KU14.** how to prepare the pipes in readiness for the marking out activities Preparation of pipes: e.g. visually checking for defects, cleaning the materials, removing burrs and sharp edges, etc.
- KU15.** selection and establishment of a suitable datum
- KU16.** importance of ensuring that marking out is undertaken from the selected datum
- KU17.** possible effects of working from an incorrect datum
- KU18.** mark-out conventions when marking out the workpiece
- KU19.** how to determine the overall length of the pipework required, taking into account allowances for pipe fittings and (where appropriate) screwed connections
- KU20.** tools and equipment used in the cutting and preparing the pipes
- KU21.** the characteristics of the various materials that are to be used with regard to the bending operations
- KU22.** need and use of heat/hot air to aid pipe bending process
- KU23.** methods used to hand bend and form the pipe
- KU24.** how to produce the various bends required
- KU25.** the reasons for incorporating expansion loops in a system, and where they should be positioned
- KU26.** range of pipe fittings that can be used, and how to identify them
- KU27.** how to produce pipe threads on the pipe ends, and the tools and
- KU28.** equipment that can be used (such as stocks and dies, pipe threading machines)
- KU29.** methods used to seal screwed joints (such as tapes and sealing compounds)
- KU30.** use of flanges to connect pipes; use of gaskets; and torque loading of flange bolts
- KU31.** methods used to prepare pipe ends and fittings for soldering or brazing, and why it is necessary to ensure that these preparations are carried out



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- KU32.** various types of soldered connectors available (such as solder ring types and capillary fittings)
- KU33.** methods used to solder the joints, and how to recognise when the fitting is correctly soldered
- KU34.** precautions to be taken when using gas torches to form the joint, and the effect of overheating the joint
- KU35.** methods used to prepare pipe ends and fittings when using adhesives, and why it is necessary to ensure that these preparations are carried out
- KU36.** methods used to cement the joints, and how to recognise when the fitting is correctly secured
- KU37.** various adhesives and sealing compounds that are used on nonmetallic pipework
- KU38.** precautions to be taken when using the adhesives, cements and sealing compounds (such as adequate ventilation, fume extraction, away from naked flames, avoiding skin contact)
- KU39.** use of compression fittings; how the pipes are sealed; and the effects of over tightening the fittings
- KU40.** use of push-fit connectors, and their advantages and disadvantages
- KU41.** how to identify the correct orientation of fittings with regard to flow, and the consequences of incorrect orientation
- KU42.** supporting methods that are used when assembling pipework, and the type of fittings that are used
- KU43.** mechanical fastenings and joining techniques: non-permanent - nuts, bolts, studs, permanent - welded, soldered, brazed, riveted
- KU44.** how to mount and secure the cutting tools in the tool holding devices
- KU45.** the methods of positioning, aligning and securing the workpiece
- KU46.** assembly methods, techniques and procedures to be used
- KU47.** methods of testing pipework systems for leaks (using air, water or hydraulic testing methods)
- KU48.** application of cutting fluids and compounds with regard to a range of different materials, and why some materials do not require cutting fluids to be used
range of materials: ferrous metals: e.g. carbon steels, stainless steels, cast iron, tool steel, hard metals; non-ferrous metals: e.g. bronze, aluminium, copper and copper alloy
- KU49.** how to check the workpiece and the measuring equipment that is used
measuring equipment: external micrometers, vernier caliper, rules, tri-squares, combination squares, protractors, feeler gauges, bore/hole gauges, radius/profile gauges, thread gauges,
- KU50.** need to check that the measuring equipment is within current calibration dates, and that the instruments are correctly zeroed
- KU51.** measuring internal and external dimensions
- KU52.** measuring geometric features
- KU53.** the importance of leaving the work area and equipment in a safe and clean condition on completion of fitting activities
- KU54.** basic rigging practice
- KU55.** different types of pipe welding methods
- KU56.** fuel brazing/soldering and cutting method



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in english and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in english and/or local language
- GS3.** convey and share technical information clearly using appropriate language
- GS4.** check and clarify task-related information
- GS5.** liaise with appropriate authorities using correct protocol
- GS6.** communicate with people in respectful form and manner in line with organizational protocol
- GS7.** undertake numerical operations, and calculations using calculators Numerical computations: addition (with decimal digits and with carrying), subtraction (with decimal digits and with borrowing), multiplication (with decimal digits), division (with decimal digit), fractions and decimals, percentages and proportions, simple ratios and averages
- GS8.** identify and draw various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder
- GS9.** demonstrate measurement and calculation of angle, perimeter, area of a common geometrical shape and can co-relate with job area requirements
- GS10.** use appropriate measuring techniques and units of measurement
- GS11.** use british and metric system of measurement and make conversions between them
- GS12.** describe the difference between celsius & fahrenheit scale and relationship between them
- GS13.** use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
- GS14.** interpret and express tolerance in terms of limits on dimensions
- GS15.** participate in on-the-job and other learning, training and development interventions and assessments
- GS16.** clarify task related information with appropriate personnel or technical adviser
- GS17.** seek to improve and modify own work practices
- GS18.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS19.** identify problems with work planning, procedures, output and behavior and their implications
- GS20.** prioritize and plan for problem solving
- GS21.** communicate problems appropriately to others
- GS22.** identify sources of information and support for problem solving
- GS23.** seek assistance and support from other sources to solve problems
- GS24.** identify effective resolution techniques
- GS25.** select and apply resolution techniques
- GS26.** seek evidence for problem resolution



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- GS27.** plan, prioritize and sequence work operations as per job requirements
- GS28.** organize and analyze information relevant to work
- GS29.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS30.** undertake and express new ideas and initiatives to others
- GS31.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS32.** ones competencies in new and different situations and contexts to achieve more
- GS33.** exercise restraint while expressing dissent and during conflict situations
- GS34.** avoid and manage distractions to be disciplined at work
- GS35.** manage own time for achieving better results
- GS36.** work in a team in order to achieve better results
- GS37.** identify and clarify work roles within a team
- GS38.** communicate and cooperate with others in the team for better results
- GS39.** seek assistance from fellow team members

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working Safely</i>	5	22	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	1	3	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (ppe) and other relevant safety regulations while performing fabrication and fitting operations	1	4	-	-
PC3. ensure work area is clean and safe from hazards Hazards: use of power tools, trailing leads or hoses, damaged or badly maintained tools and equipment; using files with damaged or poor fitting handles; using machine tools; misuses of tools; not following laid-down maintenance procedures; hazardous working condition e.g. pressurized line, high temperature area, inflammable fluid or gas lines, working at confined space or at height etc.	1	3	-	-
PC4. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	-	2	-	-
PC5. ensure that all machines and machine tools are secured at all times	-	2	-	-
PC6. work safely in trenches, confined spaces and excavated areas	-	3	-	-
PC7. observe safety measures while working on high pressure line/system (steam, compressed air, hydraulic etc.	1	3	-	-
PC8. follow warning and safety signs (danger, out of service, etc.) while working with energized system (electrical systems, steam & compressed air system etc.) including road safety signs	1	2	-	-
<i>Preparing for pipe fabrication, fitting or assembling operations</i>	4	9	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. determine job requirement from job specification documents obtained from valid sources Job requirements: raw materials or components required (type, quality, quantity); dimensions; limits and tolerances; operations required (list, sequence and procedures where applicable); shape or profiles to be fabricated; cutting, bending for fabricated forms; instruments and tools to be used; interdependencies; timelines Job specification documents: detailed component drawings; approved sketches/illustrations; national, international and organisational standards; reference tables and charts; fabrication/casting drawings Valid source: job instruction sheet/job card; work drawings and instructions; planning documentation; quality control documents; operation sheets; process specifications; instructions from supervisor	2	3	-	-
PC10. comply with health and safety, environmental and other relevant regulations and guidelines at work	1	2	-	-
PC11. obtain the appropriate equipment, parts and accessories for the pipe fitting or assembling operation	1	2	-	-
PC12. check that all measuring equipment is within calibration date Measuring equipment: external/internal micrometers, vernier caliper, tri-square, combination squares, rules, squares, protractors, depth micrometers, depth verniers, feeler gauges, bore/hole gauges, radius/profile gauges, thread gauges, hardness tester	-	2	-	-
<i>Marking out the components</i>	7	13	-	-
PC13. prepare/determine suitable datum from which to mark out	1	2	-	-
PC14. apply a marking medium to enhance clarity of the marking out	-	2	-	-
PC15. use an appropriate method of marking out Marking out methods: e.g. direct marking using tapes and markers, set-outs of pipework using templates, producing set wires, set-outs of pipework onto floor	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. use a range of marking out equipment (e.g. rules, squares, scribes, vernier instruments) Marking tools: rules/tapes, dividers/trammels, scribes, punches, scribing blocks, squares, protractor, permanent markers	2	2	-	-
PC17. mark out a range of features Features: datum lines; cutting guidelines; square and rectangular profiles; circular and radial profiles; angles; holes linearly positioned, boxed and on pitch circles	2	3	-	-
PC18. plan the pipe fitting activities before starting	1	2	-	-
<i>Performing pipe fitting operations</i>	2	8	-	-
PC19. cut the pipes to the appropriate lengths making allowances for bending using appropriate cutting operations and techniques Pipe cutting operations: cutting pipes to length with appropriate allowance for fittings, removing all external and internal burrs, cleaning pipe ends for soldering or cementing (where appropriate) or prior to aligning pipe for welding, the ends of the pipe shall be bevelled, cutting threads on pipe ends to the appropriate length (where appropriate), checking that prepared pipes are the correct length Pipe cutting techniques: cutting out the rough profile using saws (e.g. hacksaw, chop saw, power saw), or oxy-fuel gas cutting torch or using pipe/tube cutter cutting a pipe thread (e.g. tapping or dieing), de-burring reamers, rubbing with abrasive cloth, wire pipe cleaning, filing (flat, square, curved), drilling holes	1	4	-	-
PC20. produce pipework bends using the appropriate tools and equipment for the types and sizes of pipe Pipe bending tools and equipment: hand operated pipe bender, bending springs, pipe expander, swaging kit, hydraulic pipe bending equipment, heating methods, fillers Pipework bends and forms: angular bends, offsets, bridge sets, radii, internal, swaged ends, expansion loops, external swaged ends	1	4	-	-
<i>Performing assembling operations</i>	2	15	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. assemble and secure the pipework as per job specifications using appropriate pipe assembly methods and techniques pipe assembly methods: securing Pipework supports to structures, connecting pipe to equipment, fitting pipework supports, using gaskets, seals/sealing tapes or jointing compounds, connecting pipe-to-pipe, alignment/levelling equipment Techniques: compression fittings, soldered fittings, cemented fittings, snap-on/push fittings, brazed fittings, welded joints, screwed connections	1	3	-	-
PC22. produce pipework assemblies which combine a range of different fittings Pipe fittings: straight couplings, elbows, tee pieces, flanges, reduction pieces, drain/bleeding devices, unions	1	4	-	-
PC23. dismantle pipework assemblies without damage to components and/or subassemblies Methods to dismantle: procedure for isolation and locking off a device/system; sequence of operations used to dismantle a device/system; proof marking, correct storage procedures for removed parts; release of pressure/force; extraction	-	3	-	-
PC24. deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve	-	2	-	-
PC25. keep the work area in a safe and tidy condition during and on completion of the manufacturing activities	-	2	-	-
PC26. return all tools and equipment to the correct location on completion of the fitting activities	-	1	-	-
<i>Measuring and checking component</i>	5	8	-	-
PC27. perform the necessary checks for correct pipework assembly and dimensional accuracy Checks: hydraulic pressure testing, gas/air leakage test, water leakage testing Dimensions: linear dimensions (e.g. lengths, depths), diameters (e.g. external, internal), flatness, squareness, angles, profiles, hole size and position, thread size and fit	1	2	-	-
PC28. use the appropriate measuring equipment for checking activities	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. produce components within all of the applying standards:Components quality standards: pipes are bent to the appropriate shape/form andposition, all pipe bends are free from buckling or deformation, appropriate fittings are used, and are secure and leak free, soldered and cemented fittings are free from excessive residues, the completed assembly meets the specific system requirements	2	2	-	-
PC30. generate stage inspection reports.	1	2	-	-
NOS Total	25	75	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0201
NOS Name	Perform fitting and assembly operations on pipes to produce pipework systems
Sector	Power
Sub-Sector	Generation
Occupation	Thermal Power Plant Equipment Erection & Maintenance
NSQF Level	3
Credits	8.5
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026



Qualification Pack

SGJ/N1702: Optimize resource utilization at workplace

Description

This unit is about adopting sustainable practices and optimizing use of resources, especially material, energy and waste, in day-to-day operations at work

Scope

The scope covers the following :

- Material conservation practices
- Energy/electricity conservation practices
- Effective waste management/recycling practices

Elements and Performance Criteria

Material conservation practices

To be competent, the user/individual on the job must be able to:

- PC1.** identify ways to optimize usage of material including water in various tasks/activities/processes
- PC2.** check for spills/leakages in various tasks/activities/processes
- PC3.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC4.** carry out routine cleaning of tools, machines and equipment

Energy/electricity conservation practices

To be competent, the user/individual on the job must be able to:

- PC5.** identify ways to optimize usage of electricity/energy in various tasks/activities/processes
- PC6.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC7.** report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment
- PC8.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management/recycling practices

To be competent, the user/individual on the job must be able to:

- PC9.** identify recyclable and non-recyclable, and hazardous waste generated
- PC10.** segregate waste into different categories
- PC11.** dispose non-recyclable waste appropriately
- PC12.** deposit recyclable and reusable material at identified location
- PC13.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** potential hazards, risks and threats based on the nature of work
- KU2.** layout of the workstation and electrical and thermal equipment used
- KU3.** organizations procedures for minimizing waste
- KU4.** efficient and inefficient utilization of material and water
- KU5.** ways of efficiently managing material and water in the process
- KU6.** basics of electricity and prevalent energy efficient devices
- KU7.** ways to recognize common electrical problems
- KU8.** common practices of conserving electricity
- KU9.** usage of different colours of dustbins
- KU10.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU11.** waste management and methods of waste disposal
- KU12.** common sources of pollution and ways to minimize it

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** read Standard Operating Practices (SOP) documents
- GS4.** communicate with colleagues on the significance of greening of jobs
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** complete tasks efficiently and accurately within stipulated time
- GS7.** work with supervisors/team members to carry out work related tasks
- GS8.** identify cause and effect of greening of jobs

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Material conservation practices</i>	4	8	-	-
PC1. identify ways to optimize usage of material including water in various tasks/activities/processes	1	2	-	-
PC2. check for spills/leakages in various tasks/activities/processes	1	2	-	-
PC3. plug spills/leakages and escalate to appropriate authority if unable to rectify	1	2	-	-
PC4. carry out routine cleaning of tools, machines and equipment	1	2	-	-
<i>Energy/electricity conservation practices</i>	4	8	-	-
PC5. identify ways to optimize usage of electricity/energy in various tasks/activities/processes	1	2	-	-
PC6. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	1	2	-	-
PC7. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	1	2	-	-
PC8. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	2	-	-
<i>Effective waste management/recycling practices</i>	5	10	-	-
PC9. identify recyclable and non-recyclable, and hazardous waste generated	1	2	-	-
PC10. segregate waste into different categories	1	2	-	-
PC11. dispose non-recyclable waste appropriately	1	2	-	-
PC12. deposit recyclable and reusable material at identified location	1	2	-	-
PC13. follow processes specified for disposal of hazardous waste	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	13	26	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1702
NOS Name	Optimize resource utilization at workplace
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Resource Optimization
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	25/11/2021
Next Review Date	19/06/2025
NSQC Clearance Date	25/11/2021



Qualification Pack

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	2.0
Next Review Date	31/08/2026



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



Qualification Pack

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



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Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N0201.Perform fitting and assembly operations on pipes to produce pipework systems	25	75	-	-	100	60
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
SGJ/N1702.Optimize resource utilization at workplace	13	26	-	-	39	10
PSS/N1331.Apply basic health and safety practices for power related work	39	61	0	0	100	10
PSS/N1336.Working effectively with others	34	56	0	0	90	10
Total	131	248	0	0	379	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
OEE	Overall Equipment Effectiveness



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.